1.

Step	Operation	Buffer
1	P.put(1)	[1]
2	P.put(2)	[1, 2]
3	P.put(3) *start*	[1, 2]
4	C.get()	[2]
5	P.put(3) *end*	[2, 3]
6	C.get()	[3]

P.put(3) blocks when the producer tries to add the third value because the buffer already has two items, which is its maximum capacity.

2.

Thread 1 Thread 2
| |
ch_a.get() ch_b.get()
| |
ch_a is empty ch_b is empty
| |
Empty Empty

ch_a and ch_b cannot move on because both are waiting to receive a value that the other thread has not produced. One way to prevent this is to make Thread 1 send a value through ch_a before waiting to receive from ch_b. This way, Thread 1 produces a value for Thread 2 to use, and Thread 2 can then send a value through ch_b for Thread 1 to use. This can prevent a deadlock by removing circular wait.

3. An unbounded channel allows producers to keep producing, even when the buffer fills up. This can cause problems because, even though the producer doesn't have to wait, it uses a lot of memory. In PA09, objects are managed with reference counting, so when a producer allocates an object, its reference count starts at 1. Then the producer sends the object's handle through the channel, increasing the reference count to 2. After the producer releases its original reference, the count goes back to 1. This means that the buffer would continue to grow and hold references to more objects, leading to memory usage problems.